

Review

Synthetic chromosomes for 3D functional genomics: from principles to AI-guided design

 Manon Perrot^{1,2}, Jacques Serizay¹ and Romain Koszul¹


The role of genome 3D organization for fundamental chromatin processes, such as long-range promoter-activator regulatory interactions, remains ambiguous. Synthetic genomics, defined as the rational design of kilobase-scale genomic sequences and their assembly into chromosomal segments spanning up to hundreds of kb (and ultimately, entire genomes), provides powerful solutions to lift this ambiguity and directly test structure-function relationships through controlled perturbations. In this review, we highlight how synthetic genomics approaches overcome traditional experimental constraints by creating simplified but functional systems that isolate specific organizational principles, so far mostly in budding yeast and mammalian cells. Combined with AI-based sequence design, these approaches not only provide unprecedented insight into the organizational logic governing genome function but also open new avenues for therapeutic genomic engineering.

Addresses

¹ Institut Pasteur, CNRS UMR 3525, Université Paris Cité, Unité Régulation Spatiale des Génomes, Paris F-75015 France

² Sorbonne Université, Collège Doctoral, Paris, France

Corresponding author: Koszul, Romain

(romain.koszul@pasteur.fr), Koszul, Romain (X [@rkoszul.bsky.social](https://twitter.com/rkoszul))

Current Opinion in Genetics & Development 2026, 98:102471

This review comes from a themed issue on **Genome Architecture and Expression**

Edited by **Luca Giorgetti** and **Daniel Jost**

For complete overview of the section, please refer to the article collection, “[Articles from the Special Issue on Genome Architecture and Expression \(2026\)](#)”

Available online 15 April 2026

<https://doi.org/10.1016/j.gde.2026.102471>

0959–437X/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Introduction

In the last two decades, the three-dimensional (3D) organization of genomes has been shown to influence or regulate functions from gene expression, DNA replication, and cellular identity [1,2] (this issue). Separating correlation from causation has long been challenged by

the coevolution between sequence and chromatin structure, function, and the difficulty of manipulating the large genomic regions involved in these regulations. These works are now considerably facilitated by techniques allowing the precise manipulation of a genome structure or the integration of DNA molecules to decipher and disentangle the causal relationships between chromosome folding and metabolic processes. On one hand, massive parallel reporter assays (MPRAs) allow high-throughput modifications of genomic loci at discrete positions using oligonucleotides of a few hundred base pairs, providing a quantitative foundation for understanding how individual genomic elements, including enhancers and promoters, regulate transcription within a given genomic locus. For example, the transcriptional activity of a reporter gene can be measured under the control of millions of random variations of an individual promoter sequence in yeast, revealing the unexpected prevalence of random sequences acting as functional transcription factor binding sites (TFBSs) potent for transcription initiation [3]. Although powerful, MPRA approaches remain limited in their scope, since they do not probe the positional and combinatorial effects of regulatory elements across large genomic distances. On the other hand, synthetic genomics approaches, which consist of designing *in silico* long DNA molecules (from a few to hundreds of kb) and assembling them *in vivo* using a chassis strain (e.g. of *Saccharomyces cerevisiae* or *Escherichia coli*), have been rapidly emerging to manipulate much larger, chromosome-scale DNA molecules, with no limitations in the number of variations introduced. These long DNA molecules can eventually be transferred into other prokaryotic or eukaryotic species either in replacement of existing DNA sequences or as a supernumerary chromosome carrying novel, simplified, recoded, or optimized functions [4–6]. Pioneering works leading to the synthetic reproduction of viral, bacterial, and eukaryotic genomes have democratized the wider use of such large DNA segments [7–9].

We now believe that the convergence between synthetic and 3D genomics has reached a turning point, with recent works demonstrating both the interest and amenability of such approaches not only to learn about fundamental principles of genome folding, but also to take into consideration the multiscale organization of designer DNA when designing artificial sequences. In this review, we first discuss how large, chromosome-scale DNA sequences introduced into a nucleus can be used to explore

the host genome folding rules, and we then focus on how synthetic genomics can facilitate deciphering genetic regulation in the broader genome 3D organization context. We also predict that, empowered by artificial intelligence (AI), approaches to design a la carte functional chromatin regions may soon integrate 3D chromosome constraints to enable the construction of synthetic genomes that function reliably within the nuclear landscape.

Yeast as a platform for synthetic genomics and 3D functional genome exploration

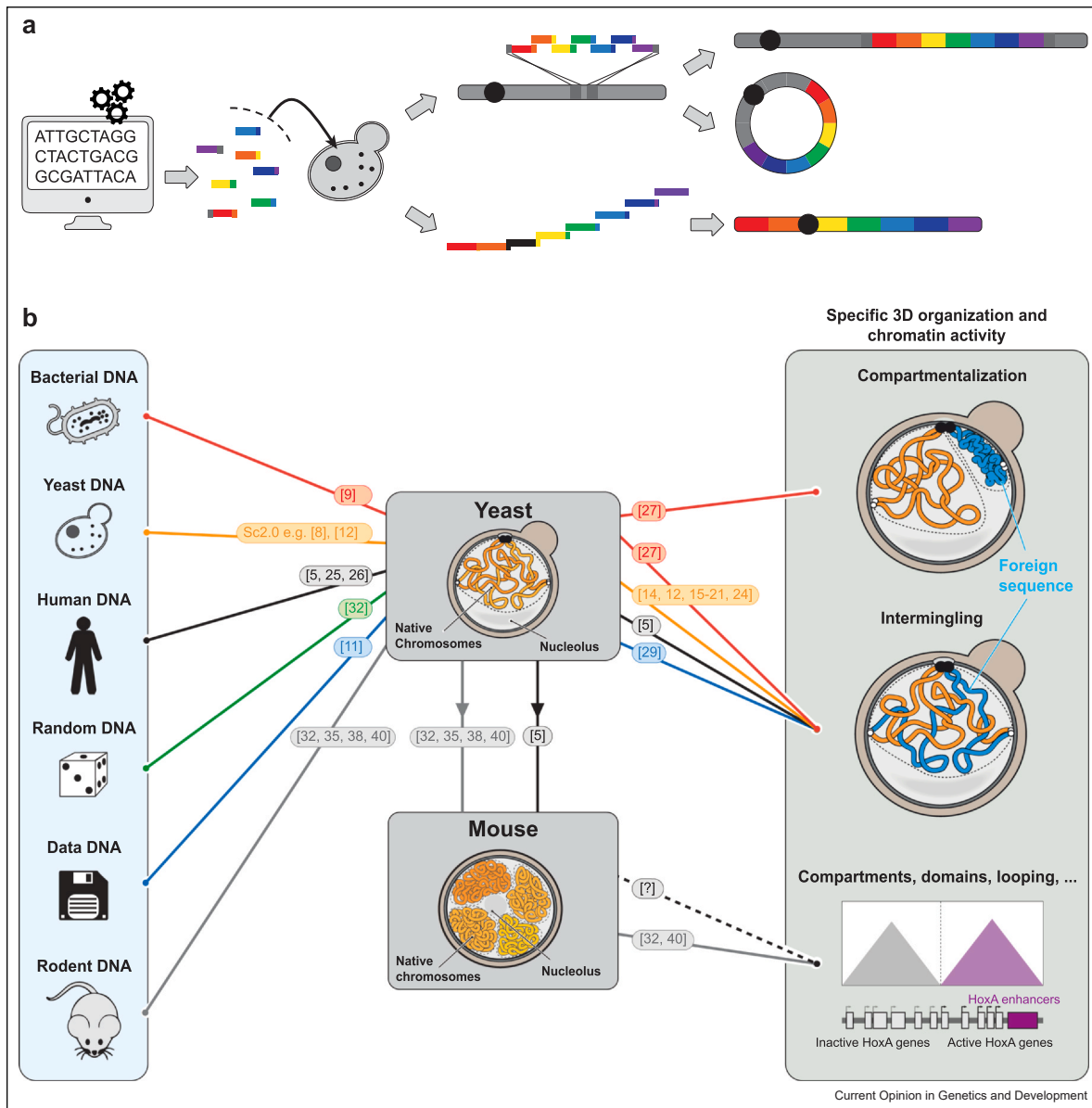
Budding yeast *S. cerevisiae*'s efficient homologous recombination machinery and remarkable resilience to long foreign DNA molecules have long made it the organism of choice to assemble self-replicating, large DNA regions used in YAC libraries [10], chromosome synthesis [9], or also DNA data storage projects [11] (Figure 1a, b). To our knowledge, the first time a large synthetic DNA segment was specifically designed to explore a question related to the 3D organization of chromosomes was in the context of studying chromosome dynamics during meiosis. In Muller et al. [12], a 144 kb synthetic chromosomal region was designed so that it could be distinguished, through polymorphism, from its homolog in an isogenic diploid strain using genomic approaches such as chromosome conformation capture (Hi-C) [13]. In this work, 3229 edits distributed across hundreds of positions were introduced in the region at once, corresponding to ~2% of all nucleotides, and highlighting the power of the approach compared to serial modifications introduced using targeted mutations. This enabled the characterization of chromosome compaction and the folding of homologues into chromatin loops at different meiotic stages, and the tracking of recombination intermediates between homologs in the context of the 3D genome [12]. This work pioneered how large synthetic DNA could be used to explore and disambiguate complex structure — function interplays (Figure 2).

Following a different path, the Sc2.0 project has sought to synthesize the entire 12 Mb *S. cerevisiae* genome into a version where transposons, introns, and tRNAs are eliminated from chromosomes, and where TAG stop codons are changed to TAA [8]. The average structures of individual synthetic Sc2.0 chromosomes are not strongly affected by these edits, as assessed by Hi-C on the first six synthetic chromosomes [14]. Each of the 16 yeast chromosomes now has its Sc2.0 counterpart, albeit in individual yeast strains, and a strain with a complete Sc2.0 karyotype is still missing. A series of works have now characterized the 3D folding of up to six and a half Sc2.0 chromosomes integrated into the same strain in the nuclear space of asynchronous cells, confirming the conservation of short-range and trans-contacts compared to wild-type ones [15–17]. A circular 186 kb neo-chromosome carrying 275 tRNA and a yeast centromere was also generated, colocalizing near the clustered yeast

centromeres [18]. Several works have further exploited *syn* chromosomes through their SCRaMbLE System, for Synthetic Chromosome Rearrangement and Modification by LoxP-mediated Evolution, which consists of the introduction of hundreds of 34 bp LoxP sites decorating *syn* chromosomes every ~3 kb. These short repeats have been harnessed in multiple ways, from Cre recombination and genome reshuffling to dCas9 targeting. For instance, LoxP sites can be targeted to the inner nuclear membrane using zinc-finger proteins designed toward LoxP [19]. This relocation encompasses neighboring regions, which become transcriptionally inactive in return [20]. In addition, engineering of *syn* chromosomes can unveil overlooked 3D patterns: the fusion between the tiny (180 kb) *synI* with *synIX* and *synIII* magnifies Sir3-dependent contacts between the mating type cassettes [21]. Also, Cre-mediated LoxP recombination frequencies strongly correlate with local Hi-C contacts (e.g. [22,23]) and open chromatin enrichment [24].

In contrast to genome rewriting projects such as Sc2.0, a more radical approach to exploring the functional 3D regulation of genomes consists of studying how chromosomes from other species, synthetically assembled in *S. cerevisiae*, fold in the yeast's nucleus. These sequences have not coevolved and have not been tuned to the metabolic rules of their host, and can be used to shed new light on these rules. The lack of heterochromatin in *S. cerevisiae* correlates with a remarkably permissive chromatin landscape, where >99% of exogenous sequences of GC range ranging from 40% to 50% (compared to *S. cerevisiae* 38%) introduced into the genome are transcribed, whether they are of mammalian [25,26], bacteria [27], random [28] or data-encoded origin [11,26]. In [29], the authors used Hi-C to show that, in asynchronous cells, the transcribed 254 kb data chromosome of [11] mingled with the native genome with no particular pattern (Figure 1b). Meneu et al. further showed that a 1 Mb-long, 40% GC bacterial chromosome introduced in the yeast nucleus mingles during G1 with native chromosomes, but exhibits longer G2/M cohesin-mediated loops and longer transcription tracks [27]. In contrast, extreme sequence composition can override *S. cerevisiae* permissive default state, as AT-rich (24% GC) chromosomes introduced into its genome are not transcribed, show longer linker DNA, and physically self-associate, segregating from active chromosomes into an inactive compartment. The resulting active/inactive compartmentalization in the absence of canonical heterochromatin points at alternative mechanisms driving compartment formation, suggesting that fundamental principles underlying genome organization have yet to be elucidated [27]. Compartmentalization of synthetic sequences disappears during G2/M when DNA is folded into loops by cohesin, similarly to mammalian mitotic chromosome compaction [27]. This inactive chromosomal compartment displays further remarkable properties, as it titrates the 2 μ plasmid, one of the two known instances of

Figure 1



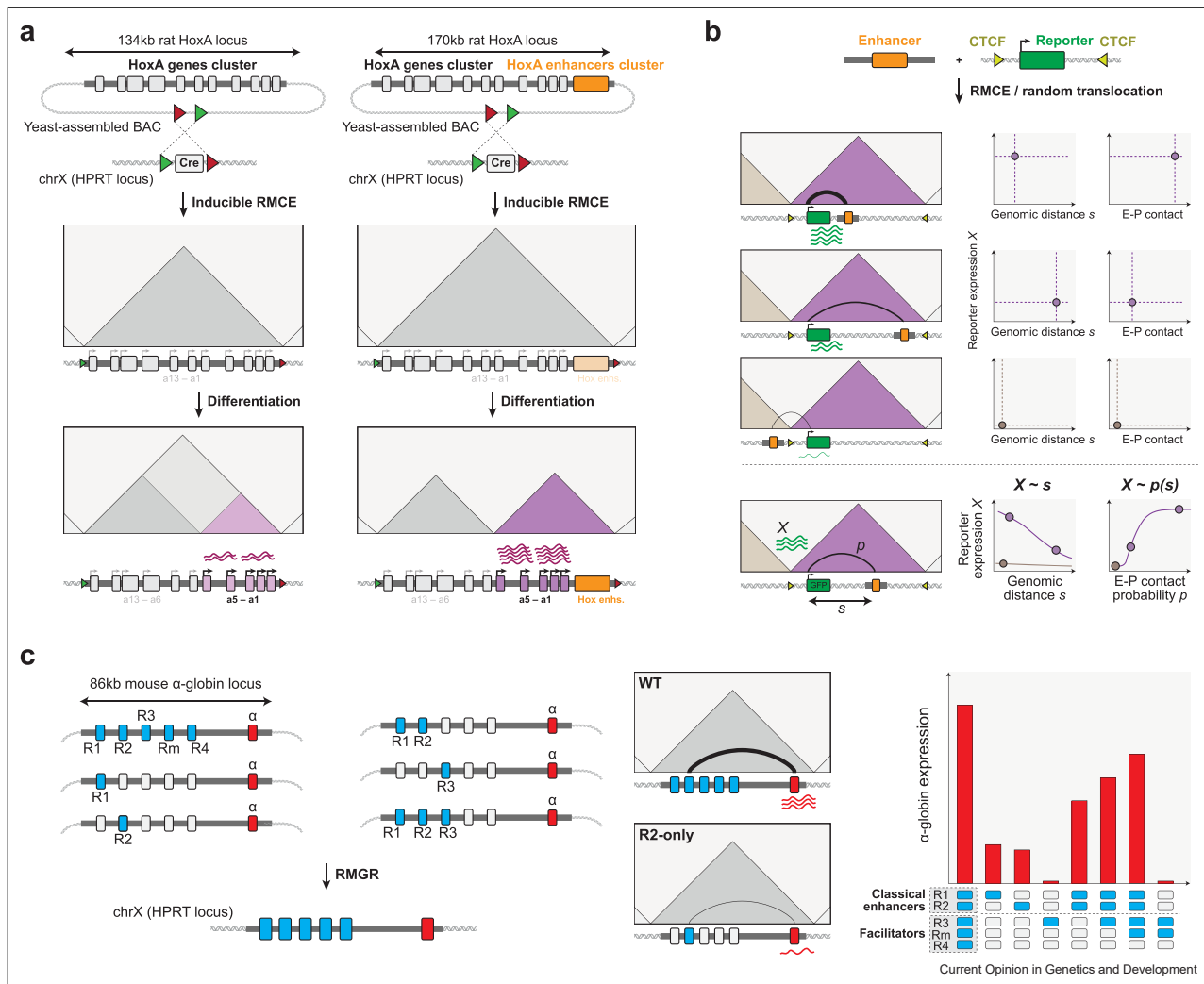
Overview of synthetic genomics projects with a focus on genome 3D architecture. **(a)** A DNA molecule is designed *in silico* to meet with the exigence of the study. Most DNA assemblies are done in budding yeast using kb-long DNA blocks (colored boxes), resulting either in replacement of the native sequence with the designer one (top), or in the addition of DNA to a native chromosome or a plasmid. These sequences can be characterized directly in yeast and/or transferred to other hosts such as mice. **(b)** Non-exhaustive lists of projects where designer DNA sequences of various origins are assembled into budding yeast *S. cerevisiae* (left references). The nature of the sequences vary, from slightly modified native sequence (e.g. meiosis chromosomes from [17]) to more dramatic changes such as the removal of one or more codons from the coding sequences (e.g. Sc2.0 [10,12]) or CRE reorganization (e.g. [44]). Some of these projects, as well as others, explored specifically the 3D organization of these designer molecules (right references).

an eukaryotic plasmid. In a strain carrying both the 2 μ and the AT-rich chromosome, the plasmid and bacterial DNA colocalize into a silent compartment (Figure 1b) [30]. Together, these synthetic DNA experiments demonstrate that genome functional organization emerges from the interplay between intrinsic sequence properties, species-specific default chromatin states, and evolutionary selection.

Synthetic 'regulatory genomics' to decode the principles of chromatin regulatory contacts

While the chromosome-scale synthetic approaches described above can be used to investigate chromosomal properties shaping their folding and compartmentalization, understanding gene regulation also requires dissecting how individual cis-regulatory elements (CREs) can interact

Figure 2



Structure-function interplay projects exploiting large-scale DNA synthesis. **(a)** Pinglay et al. synthesized the rat HoxA gene cluster and inserted it into mESCs at the HPRT TAD using RMCE. Upon differentiation, HoxA genes a1 to a5 of the synthetic locus are up-regulated, and a subTAD segregating those active genes from a6 to a13 inactive HoxA genes emerges. Up-regulation and subTAD segregation are increased when HoxA enhancers are also introduced in the synthetic locus (right). **(b)** Both Zuin et al. and Rinzema et al. investigated the role of genomic distance between an enhancer and the promoter it regulates. Using different strategies, they used a reporter gene located within a TAD with minimal regulatory activity, and integrated an enhancer at different positions within or outside the TAD. They observed a decreasing regulatory effect with increasing E-P genome distances, and a non-linear relationship between E-P contact probability and expression of the reporter gene. **(c)** Blayney et al. designed synthetic versions of the mouse α -globin locus and systematically removed individual or combinations of regulatory DHSs. They found that only 2 (R1, R2) out of the 5 DHSs had enhancer activity for α -globin expression. The remaining 3 DHSs (R3, Rm, R4) act as facilitators to increase R1/R2 enhancer activity, possibly through increase of E-P contacts, but do not bear any canonical enhancer activity on their own. mESCs: mouse embryonic stem cells; TAD: topologically associating domain; RMCE: recombinase-mediated cassette exchange; RMGR: recombinase-mediated genomic replacement; E-P: enhancer-promoter; DHS: DNase-hypersensitivity sites.

combinatorially within their local chromatin context. MPRA have provided valuable insights into the activity of isolated CREs, but remain limited in their ability to probe the positional and combinatorial effects of multiple regulatory elements across large genomic distances [31]. To address this challenge, synthetic genomics frameworks have been leveraged to precisely engineer complex regulatory regions containing defined combinations of enhancers and/or promoters, enabling a bottom-up

exploration of spatial and functional interactions between CREs that shape gene expression. Pinglay et al. synthetically reconstructed in yeast the 130 kb rat HoxA cluster and its CREs, then introduced it into the mouse genome [32]. When integrated at an ectopic genomic location, this locus was remodeled following enhancer activation to form two separate topologically associating domains (TADs [1]), one containing distal enhancers and the induced Hox genes a1 to a5, and one containing the other silent HoxA

genes (Figure 2a). Zuin et al. [33] focused on Sox2 control region (SCR), a collection of distal Sox2 enhancers. Using a transposase-based approach, the authors generated hundreds of different cell lines, each with an ectopic SCR integrated at varying distances (± 300 kb) relative to a transgenic Sox2 copy. They confirmed that while enhancer-induced Sox2 transcription decays with increasing enhancer-promoter (E-P) distances and is limited by TAD boundaries, the orientation of the SCR was irrelevant for its regulatory activity. More importantly, they revealed a non-linear relationship between SCR–Sox2 physical interactions and the SCR's regulatory activity (Figure 2b). Rinzema et al. [34] also observed this non-linear relationship when studying synthetic, minimized versions of the β -globin locus control region (LCR) (Figure 2b).

Leveraging the Big-IN genome editing platform, Brosh et al. [35] replaced the endogenous Sox2 SCR with different 40–50 kb long synthetic versions carrying deletions, inversions, or rearrangements of CREs. They found that some enhancers function autonomously while others are context-dependent, requiring neighboring elements for full activity. These studies resonate with Jensen et al., who used multiplexed arrays of guide RNAs to efficiently recruit the activator dCas9-VPR fusion protein (i.e. dCas9 fused to transcriptional activation domains VP64, p65 and Rta) to different genomic regions, demonstrating that the distance-dependent decay of enhancer-mediated gene activation is gene-specific [36]. Thomas et al. [37] further elucidated principles of E-P regulatory interactions by engineering different synthetic versions of the regulatory landscape around the hemoglobin subunit beta (Hbb) locus to demonstrate that distal enhancers can act in synergy to induce enhancer-mediated gene activation, regardless of their individual strength and orientation. Weak enhancers located between a strong distal enhancer and a promoter significantly increased gene induction, suggesting that enhancers can compensate for each other and cooperate over large genomic distances. Reusing the Big-IN framework, Ordoñez et al. [38,39] confirmed these observations by characterizing the activity of individual CREs of synthetic loci containing Sox2 and different combinations of manually selected enhancers from the Sox2 endogenous SCR or from other genomic loci. They also observed context-dependent activity and cooperativity between enhancers, showing that this cooperativity between neighboring DHSs decays as a function of distance, extending up to several kilobases. Testing a number of different arrangements of CREs, Blayney et al. [40] concluded that enhancers physically in contact with 'facilitators' sequences led to increased transcriptional induction of their target (Figure 2c). LCRs thus exhibit emergent properties resulting from interactions between classical enhancers and other regulatory loci, with these interactions constrained by chromatin topology. This model of flexible and

synergistic enhancer function refined conclusions from earlier reports that adapted MPRA assays to study E-P functional interactions [41,42]. In particular, by generating a combinatorial library of thousands of enhancer-promoter (E-P) synthetic pairs and testing their regulatory potential by MPRA, Bergman et al. and Martinez-Ara et al. found that (1) the E-P regulatory activity was relatively independent of the orientation or position of the CREs, and that (2) enhancers exhibited high selectivity for the promoters they can regulate, revealing an important constraint on enhancer-promoter compatibility that shapes regulatory specificity [41,42].

Thus, in addition to their application to chromosome-scale genomic design, synthetic genomics combined with functional assays unlock the ability to decipher the cooperative, context-dependent, and spatially coordinated nature of enhancer-based gene regulation, ultimately helping to elucidate how disruptions in chromatin architecture can lead to developmental disorders and disease. Collectively, these findings support the model whereby chromatin loops physically bring distant regulatory elements closer together and allow them to function specifically and cooperatively.

Toward artificial intelligence-guided synthetic chromosome design

Breakthrough innovations in artificial intelligence (AI) applied to genomics and epigenomics have transformed our ability to design genomic sequences with predictable behavior. Modern deep-learning frameworks, including convolutional and attention-based neural networks (NN), have enabled the accurate prediction of the complex multi-layer organization of DNA sequence features (reviewed in [43]). A growing number of NNs have emerged to predict organizational features of a bare DNA sequence, from nucleosome positioning [44] and post-translational modifications to recruitment of transcription [45] and structural [31] factors, regulatory activity of individual CREs [46], gene expression [47], and transcript splicing [48]. When applied to synthetic sequences introduced in various organisms, NN models have been demonstrated to accurately predict these modalities [27,49,50]. Dozens of AI models now exist to predict chromatin 3D folding from DNA sequences (recently reviewed in [51]). Some (e.g. Orca [52]) capture sequence dependencies underlying compartments, TADs, and cell-type-specific interactions while others (e.g. DeepPhiC [53]) focus on predicting enhancer-promoter interactions. Ultimately, these models enable *in silico* evaluation of chromatin folding and structural variant effect prediction before synthesis.

The field of predictable biology is now gradually shifting toward programmable biology [54], using generative AI approaches to design genomic sequences *in silico*. These

synthetic sequences can encode CREs or functional genomic loci with a pre-established organization and context-dependent activity, based on implicit [55–57] or explicit chromatin features [50,58]. At a larger scale, chromosome-long synthetic sequences and even full viral and prokaryotic genomes can be designed using generative AI [59–61]. Enformer [45] or Chromoformer [62] models use Hi-C data as input to predict the multi-modal epigenomic landscape of natural or synthetic sequences, but they are not generative models. Thus, even though spatial chromatin folding plays a direct role in gene regulation and chromosome metabolism, current approaches fail to integrate 3D genome organization principles into the generative design process itself.

Conclusion

By enabling systematic manipulation and reconstruction of regulatory landscapes at scales ranging from individual elements to entire gene clusters, synthetic genomics approaches are transforming our understanding of gene regulation from a catalog of individual parts to a mechanistic framework of 3D spatially organized, co-operative regulatory systems. In parallel, integrating large synthetic DNA molecules into host genomes has delivered new perspectives on fundamental chromatin processes such as compaction, transcription, or compartmentalization, and has revealed how intrinsic sequence properties dictate species-specific chromatin states to shape genome organization. Altogether, these insights not only illuminate fundamental principles of genome organization but also provide a roadmap for rational design of synthetic regulatory circuits with predictable transcriptional outputs. Beyond these conceptual advances, the remarkable modularity of synthetic genomics fuels innovation in experimental design, with original work, such as that of Lu et al. (pre-print), constantly highlighting its broader creative potential not only for genome refactoring but also for the construction of reusable, hierarchical design frameworks. Note that synthetic chromosome assembly still faces both experimental and computational hurdles. The synthesis and assembly of long, complex DNA sequences remain constrained by specific sequence properties, such as extreme GC content or repetitive motifs. Maintaining large engineered constructs stably embedded within dynamic chromatin environments can also prove challenging. In addition, predictive computational tools remain heavily dependent on their training datasets, which can limit their ability to accurately model sequences that substantially deviate from natural genomes [63].

The convergence of synthetic genomics and artificial intelligence opens unprecedented opportunities for designing chromosomes with specific architectural features. Still, existing generative models predominantly rely on

linear sequence features and 1D epigenomic annotations, essentially treating the genome as a string of regulatory elements rather than as a spatially organized polymer with distance-dependent interactions, loop domains, and topological constraints. This represents a critical gap: the synthetic genomics studies reviewed here underscore that the function of regulatory elements is profoundly shaped by spatial context, including enhancer-promoter distances, TAD boundaries, and the presence of architectural elements like CTCF sites, yet these spatial design rules remain largely absent from current AI-driven genome design frameworks. Moreover, the causal relationships between sequence features and 3D structure remain incompletely understood, making it challenging to specify which sequence modifications will reliably produce desired spatial configurations. Successfully designing functional regulatory genomic sequences will therefore require hybrid approaches that combine generative sequence models with explicit 3D structure prediction and optimization, enabling the rational design of synthetic loci where both the regulatory sequence content and the resulting chromatin architecture can be specified and validated.

Past achievements in engineering artificial yeast and human chromosomes demonstrate that understanding the 3D architecture of the genome is an essential step in creating functional synthetic chromosomes [64], requiring the integration of chromatin biology, nuclear organization, and evolutionary principles. The intrinsic self-organizing properties of chromatin suggest that artificial chromosomes can exploit robust organizational principles: future chromosome designs could incorporate AI-optimized regulatory elements and boundary elements to introduce predictable folding patterns, phase separation, or novel nuclear compartments. Such *a la carte* chromatin domains would not only enable the construction of synthetic genomes with reliable function, but also provide powerful tools to finely dissect how chromatin composition and high-order structure influence DNA-dependent processes such as replication, recombination, or repair [65,66]. This convergence of synthetic genomics, 3D functional genomics, and artificial intelligence represents a new era in which the spatial organization of chromosomes is designed with as much precision as the DNA sequence itself.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could

have appeared to influence the work reported in this paper.

Acknowledgements

This work was funded, in whole or in part, by grants from the French government, managed by the Agence Nationale de la Recherche under the France 2030 programme to R.K. [ANR-23-CHBS-0002; ANR-22-CE12-0015]. A CC-BY public copyright license has been applied by the authors to the present document and on all subsequent versions up to the author-accepted manuscript version arising from this submission, in accordance with the grants' open-access conditions.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. Dekker J, Mirny LA: **The chromosome folding problem and how cells solve it.** *Cell* 2024, **187**:6424-6450.
 2. Szalay M-F, Majchrzycka B, Jerković I, Cavalli G, Ibrahim DM: **Evolution and function of chromatin domains across the tree of life.** *Nat Struct Mol Biol* 2024, **31**:1824-1837.
 3. de Boer CG, Vaishnav ED, Sadeh R, Abeyta EL, Friedman N, Regev A: **Deciphering eukaryotic gene-regulatory logic with 100 million random promoters.** *Nat Biotechnol* 2020, **38**:56-65.
 4. Galanie S, Thodey K, Trenchard IJ, Filsinger Interrante M, Smolke CD: **Complete biosynthesis of opioids in yeast.** *Science* 2015, **349**:1095-1100.
 5. Liu Y, Zhou J, Liu D, Hu X, Yang L, Song X-R, Jin X-D, Xie W, Yang L, Liu Z, *et al.*: **De novo assembly and delivery of synthetic megabase-scale human DNA into mouse early embryos.** *Nat Methods* 2025, **22**:1686-1697.
 6. Robertson WE, Rehm FBH, Spinck M, Schumann RL, Tian R, Liu W, Gu Y, Kleefeldt AA, Day CF, Liu KC, *et al.*: **Escherichia coli with a 57-codon genetic code.** *Science* 2025, **390**:eady4368.
- The authors synthesize a functional 4-Mb *E. coli* genome (Syn57) with radical genetic code compression, replacing all occurrences of seven different codons with synonyms to create a 57-codon organism. This landmark work established an iterative design-build-test framework for synthetic genomics and is a foundation for engineering orthogonal genetic systems.
7. Cello J, Paul AV, Wimmer E: **Chemical synthesis of poliovirus cDNA: generation of infectious virus in the absence of natural template.** *Science* 2002, **297**:1016-1018.
 8. Dymond JS, Richardson SM, Coombes CE, Babatz T, Muller H, Annaluru N, Blake WJ, Schwerzmann JW, Dai J, Lindstrom DL, *et al.*: **Synthetic chromosome arms function in yeast and generate phenotypic diversity by design.** *Nature* 2011, **477**:471-476.
 9. Gibson DG, Benders GA, Andrews-Pfannkoch C, Denisova EA, Baden-Tillson H, Zaveri J, Stockwell TB, Brownley A, Thomas DW, Algire MA, *et al.*: **Complete chemical synthesis, assembly, and cloning of a *Mycoplasma genitalium* genome.** *Science* 2008, **319**:1215-1220.
 10. Burke DT, Carle GF, Olson MV: **Cloning of large segments of exogenous DNA into yeast by means of artificial chromosome vectors.** *Science* 1987, **236**:806-812.
 11. Chen W, Han M, Zhou J, Ge Q, Wang P, Zhang X, Zhu S, Song L, Yuan Y: **An artificial chromosome for data storage.** *Natl Sci Rev* 2021, **8**:nwab028.
 12. Muller H, Scolari VF, Agier N, Piazza A, Thierry A, Mercy G, Descorps-Declere S, Lazar-Stefanita L, Espeli O, Llorente B, *et al.*: **Characterizing meiotic chromosomes' structure and pairing using a designer sequence optimized for Hi-C.** *Mol Syst Biol* 2018, **14**:e8293.
 13. Lieberman-Aiden E, Berkum NL van, Williams L, Imakaev M, Ragoczy T, Telling A, Amit I, Lajoie BR, Sabo PJ, Dorschner MO, *et al.*: **Comprehensive mapping of long-range interactions reveals folding principles of the human genome.** *Science* 2009, **326**:289-293.
 14. Mercy G, Mozziconacci J, Scolari VF, Yang K, Zhao G, Thierry A, Luo Y, Mitchell LA, Shen M, Shen Y, *et al.*: **3D organization of synthetic and scrambled chromosomes.** *Science* 2017, **355**:eaaf4597.
 15. Lauer S, Luo J, Lazar-Stefanita L, Zhang W, McCulloch LH, Fanfani V, Lobzaev E, Haase MAB, Easo N, Zhao Y, *et al.*: **Context-dependent neocentromere activity in synthetic yeast chromosome VIII.** *Cell Genom* 2023, **3**:100437.
 16. Shen Y, Gao F, Wang Y, Wang Y, Zheng J, Gong J, Zhang J, Luo Z, Schindler D, Deng Y, *et al.*: **Dissecting aneuploidy phenotypes by constructing Sc2.0 chromosome VII and SCRaMbLEing synthetic disomic yeast.** *Cell Genom* 2023, **3**:100364.
 17. Zhao Y, Coelho C, Hughes AL, Lazar-Stefanita L, Yang S, Brooks AN, Walker RSK, Zhang W, Lauer S, Hernandez C, *et al.*: **Debugging and consolidating multiple synthetic chromosomes reveals combinatorial genetic interactions.** *Cell* 2023, **186**:5220-5236.e16.
 18. Schindler D, Walker RSK, Jiang S, Brooks AN, Wang Y, Müller CA, Cockram C, Luo Y, Garcia A, Schraivogel D, *et al.*: **Design, construction, and functional characterization of a tRNA neochromosome in yeast.** *Cell* 2023, **186**:5237-5253.e22.
 19. Ichikawa DM, Abidin O, Alerasool N, Kogenaru M, Mueller AL, Wen H, Giganti DO, Goldberg GW, Adams S, Spencer JM, *et al.*: **A universal deep-learning model for zinc finger design enables transcription factor reprogramming.** *Nat Biotechnol* 2023, **41**:1117-1129.
 20. Zhang W, Lazar-Stefanita L, Yamashita H, Shen MJ, Mitchell LA, Kurasawa H, Lobzaev E, Fanfani V, Haase MAB, Sun X, *et al.*: **Manipulating the 3D organization of the largest synthetic yeast chromosome.** *Mol Cell* 2023, **83**:4424-4437.e5.
 21. Luo J, Vale-Silva LA, Raghavan AR, Mercy G, Heldrich J, Sun X, Li MK, Zhang W, Agmon N, Yang K, *et al.*: **Synthetic chromosome fusion: effects on mitotic and meiotic genome structure and function.** *Cell Genom* 2023, **3**:100439.
 22. Burgess SM, Kleckner N: **Collisions between yeast chromosomal loci in vivo are governed by three layers of organization.** *Genes Dev* 1999, **13**:1871-1883.
- The Sc2.0 yeast synthetic genome project has seen recent and continuous progress, including here the construction of a fusion chromosome of multiple synthetic chromosomes (synI-synIII-synIX), the de novo synthesis of a 190kb-long tRNA neochromosome housing all 275 relocated nuclear tRNA genes, and the silencing of the entire synthetic synIV chromosome by artificially relocating it to the internal nuclear membrane. These designer chromosomes revealed principles of genome organization, including the role of nuclear positioning for gene activity in yeast, demonstrating how synthetic genomics can deconvolute complex biological systems and test hypotheses about essential genome functions.
23. Liou VS, Cournac A, Marbouty M, Duigou S, Mozziconacci J, Espéli O, Boccard F, Koszul R: **Multiscale structuring of the *E. coli* chromosome by nucleoid-associated and condensin proteins.** *Cell* 2018, **172**:771-783.e18.
 24. Zhou S, Wu Y, Zhao Y, Zhang Z, Jiang L, Liu L, Zhang Y, Tang J, Yuan Y-J: **Dynamics of synthetic yeast chromosome evolution shaped by hierarchical chromatin organization.** *Natl Sci Rev* 2023, **10**:nwad073.
- The Sc2.0 yeast synthetic genome project has seen recent and continuous progress, including here the construction of a fusion chromosome of multiple synthetic chromosomes (synI-synIII-synIX), the de novo synthesis of a 190kb-long tRNA neochromosome housing all 275 relocated nuclear tRNA genes, and the silencing of the entire synthetic synIV chromosome by artificially relocating it to the internal nuclear membrane. These designer chromosomes revealed principles of genome organization, including the role of nuclear positioning for gene activity in yeast, demonstrating how synthetic genomics can deconvolute complex biological systems and test hypotheses about essential genome functions.

25. Camellato BR, Brosh R, Ashe HJ, Maurano MT, Boeke JD:

- **Synthetic reversed sequences reveal default genomic states.** *Nature* 2024, **628**:373-380.

The Sc2.0 yeast synthetic genome project has seen recent and continuous progress, including here the construction of a fusion chromosome of multiple synthetic chromosomes (synI-synIII-synIX), the de novo synthesis of a 190kb-long tRNA neochromosome housing all 275 relocated nuclear tRNA genes, and the silencing of the entire synthetic synIV chromosome by artificially relocating it to the internal nuclear membrane. These designer chromosomes revealed principles of genome organization, including the role of nuclear positioning for gene activity in yeast, demonstrating how synthetic genomics can deconvolute complex biological systems and test hypotheses about essential genome functions.

26. Luthra I, Jensen C, Chen XE, Salaudeen AL, Rafi AM, de Boer CG: **Regulatory activity is the default DNA state in eukaryotes.** *Nat Struct Mol Biol* 2024, **31**:559-567.

27. Meneu L, Chapard C, Serizay J, Westbrook A, Routhier E, Ruault M, •• Perrot M, Minakakis A, Girard F, Bignaud A, et al.: **Sequence-dependent activity and compartmentalization of foreign DNA in a eukaryotic nucleus.** *Science* 2025, **387**:eadm9466.

The authors characterized two Mb-scale bacterial chromosomes with highly different GC% content assembled into *S. cerevisiae*, showing that the sequence composition deterministically drives transcriptional activity and 3D spatial organization. They demonstrate that euchromatin-heterochromatin-like compartmentalization can emerge spontaneously from sequence composition in a species naturally lacking such structures. They also showed that deep learning models can predict nucleosome positioning, as well as RNA Pol II and cohesin deposition along a foreign or synthetic DNA molecule from its sequence alone, highlighting sequence composition as a fundamental design parameter for engineering spatial genome organization.

28. Gvozdenov Z, Barcutean Z, Struhl K: **Functional analysis of a random-sequence chromosome reveals a high level and the molecular nature of transcriptional noise in yeast cells.** *Mol Cell* 2023, **83**:1786-1797.e5.

29. Zhou J, Zhang C, Wei R, Han M, Wang S, Yang K, Zhang L, Chen W, Wen M, Li C, et al.: **Exogenous artificial DNA forms chromatin structure with active transcription in yeast.** *Sci China Life Sci* 2022, **65**:851-860.

30. Girard F, Even A, Thierry A, Ruault M, Meneu L, Larrous P, Garnier M, Adiba S, Taddei A, Koszul R, et al.: **Parasitic plasmids are anchored to inactive regions of eukaryotic chromosomes through a nucleosome signal.** *EMBO J* 2025, **44**:2134-2156.

31. Kim S, Wysocka J: **Deciphering the multi-scale, quantitative cis-regulatory code.** *Mol Cell* 2023, **83**:373-392.

32. Pinglay S, Bulajić M, Rahe DP, Huang E, Brosh R, Mamrak NE, King BR, German S, Cadley JA, Rieber L, et al.: **Synthetic regulatory reconstitution reveals principles of mammalian Hox cluster regulation.** *Science* 2022, **377**:eabk2820.

33. Zuin J, Roth G, Zhan Y, Cramard J, Redolfi J, Piskadlo E, Mach P, Kryzhanovska M, Tihanyi G, Kohler H, et al.: **Nonlinear control of transcription through enhancer-promoter interactions.** *Nature* 2022, **604**:571-577.

34. Rinzema NJ, Sofiadis K, Tjalsma SJD, Verstegen MJAM, Oz Y, Valdes-Quezada C, Felder A-K, Filipovska T, van der Elst S, de Andrade dos Ramos Z, et al.: **Building regulatory landscapes reveals that an enhancer can recruit cohesin to create contact domains, engage CTCF sites and activate distant genes.** *Nat Struct Mol Biol* 2022, **29**:563-574.

35. Brosh R, Coelho C, Ribeiro-dos-Santos AM, Ellis G, Hogan MS, Ashe HJ, Somogyi N, Ordoñez R, Luther RD, Huang E, et al.: **Synthetic regulatory genomics uncovers enhancer context dependence at the Sox2 locus.** *Mol Cell* 2023, **83**:1140-1152.e7.

36. Jensen CL, Chen L-F, Swigut T, Crocker OJ, Yao D, Bassik MC, Ferrell JE, Boettiger AN, Wysocka J: **Long-range regulation of transcription scales with genomic distance in a gene-specific manner.** *Mol Cell* 2025, **85**:347-361.e7.

37. Thomas HF, Feng S, Haslhofer F, Huber M, Gallardo MG, Loubiere V, Vanina D, Pitasi M, Stark A, Buecker C: **Enhancer cooperativity can compensate for loss of activity over large genomic distances.** *Mol Cell* 2025, **85**:362-375.e9.

The authors developed a synthetic platform in mouse embryonic stem cells to build regulatory landscapes from the bottom up. They systematically tested how combinations of enhancers with different strengths affect transcriptional output across varying genomic distances. They discovered that weak enhancers positioned between strong distal enhancers and promoters dramatically increase gene expression and extend the functional range of enhancer action, demonstrating that enhancer synergy is distance-dependent. This highlights that synthetic regulatory design must account for both individual element properties and their spatial arrangement to achieve predictable gene expression outcomes.

38. Ordoñez R, Zhang W, Ellis G, Zhu Y, Ashe HJ, Ribeiro-dos-Santos AM, Brosh R, Huang E, Hogan MS, Boeke JD, et al.: **Genomic context sensitizes regulatory elements to genetic disruption.** *Mol Cell* 2024, **84**:1842-1854.e7.

39. Ordoñez R, Ribeiro-dos-Santos AM, McLoughlin C, Ellis G, Ashe HJ, Zufiaurre NB, Brosh R, Majewski M, Camellato B, Boeke JD, et al.: **Synthetic genomic dissection of enhancer context sensitivity and synergy.** *BioRxiv* 2025, <https://doi.org/10.1101/2025.08.13.669251>

40. Blayney JW, Francis H, Rampasekova A, Camellato B, Mitchell L, •• Stolper R, Cornell L, Babbs C, Boeke JD, Higgs DR, et al.: **Super-enhancers include classical enhancers and facilitators to fully activate gene expression.** *Cell* 2023, **186**:5826-5839.e18.

By generating a mouse model with engineered, enhancer-less α -globin regulatory locus from the bottom-up, the authors showed that super-enhancers comprise two functionally distinct element types important for *in vivo* gene regulation: classical enhancers with intrinsic transcriptional activity and 'facilitator' elements that lack autonomous activity but are essential for full enhancer function. This work reveals that facilitators with specific DNA sequence signatures potentiate classical enhancers by enabling Mediator recruitment and E-P interactions. These results emphasize that the synthetic design of regulatory DNA sequences requires understanding not only the activity of individual CREs but also the synergistic roles of non-canonical CREs within regulatory clusters.

41. Bergman DT, Jones TR, Liu V, Ray J, Jagoda E, Siraj L, Kang HY, Nasser J, Kane M, Rios A, et al.: **Compatibility rules of human enhancer and promoter sequences.** *Nature* 2022, **607**:176-184.

42. Martinez-Ara M, Comoglio F, van Arensbergen J, van Steensel B: **Systematic analysis of intrinsic enhancer-promoter compatibility in the mouse genome.** *Mol Cell* 2022, **82**:2519-2531.e6.

43. Barbadilla-Martínez L, Klaassen N, van Steensel B, de Ridder J: **Predicting gene expression from DNA sequence using deep learning models.** *Nat Rev Genet* 2025, **26**:666-680.

44. Routhier E, Pierre E, Khodabandelou G, Mozziconacci J: **Genome-wide prediction of DNA mutation effect on nucleosome positions for yeast synthetic genomics.** *Genome Res* 2020, **31**:317-326.

45. Avsec Ž, Weilert M, Shrikumar A, Krueger S, Alexandari A, Dalal K, Fropf R, McAnany C, Gagneur J, Kundaje A, et al.: **Base-resolution models of transcription-factor binding reveal soft motif syntax.** *Nat Genet* 2021, **53**:354-366.

46. Kelley DR, Reshef YA, Bileschi M, Belanger D, McLean CY, Snoek J: **Sequential regulatory activity prediction across chromosomes with convolutional neural networks.** *Genome Res* 2018, **28**:739-750.

47. Linder J, Srivastava D, Yuan H, Agarwal V, Kelley DR: **Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation.** *Nat Genet* 2025, **57**:949-961.

48. Jaganathan K, Kyriazopoulou Panagiotopoulou S, McRae JF, Darbandi SF, Knowles D, Li YI, Kosmicki JA, Arbelaez J, Cui W, Schwartz GB, et al.: **Predicting splicing from primary sequence with deep learning.** *Cell* 2019, **176**:535-548.e24.

49. Vaishnav ED: **The evolution, evolvability and engineering of gene regulatory DNA.** *Nature* 2022, **603**:455-463.

50. LaFleur TL, Hossain A, Salis HM: **Automated model-predictive design of synthetic promoters to control transcriptional profiles in bacteria.** *Nat Commun* 2022, **13**:5159.

51. Wall BPG, Nguyen M, Harrell JC, Dozmorov MG: **Machine and deep learning methods for predicting 3D genome organization.** *Methods Mol Biol Clifton NJ* 2025, **2856**:357-400.
 52. Zhou J: **Sequence-based modeling of three-dimensional genome architecture from kilobase to chromosome scale.** *Nat Genet* 2022, **54**:725-734.
 53. Agarwal A, Chen L: **DeepPHiC: predicting promoter-centered chromatin interactions using a novel deep learning approach.** *Bioinformatics* 2023, **39**:btac801.
 54. Abudayyeh OO, Gootenberg JS: **Programmable biology through artificial intelligence: from nucleic acids to proteins to cells.** *Nat Methods* 2024, **21**:1384-1386.
 55. Li Z, Zhang Y, Peng B, Qin S, Zhang Q, Chen Y, Chen C, Bao Y, Zhu Y, Hong Y, et al.: **A novel interpretable deep learning-based computational framework designed synthetic enhancers with broad cross-species activity.** *Nucleic Acids Res* 2024, **52**:13447-13468.
 56. Gosai SJ, Castro RI, Fuentes N, Butts JC, Mouri K, Alasoadura M, Kales S, Nguyen TTL, Noche RR, Rao AS, et al.: **Machine-guided design of cell-type-targeting cis-regulatory elements.** *Nature* 2024, **634**:1211-1220.
 57. DaSilva LF, Senan S, Patel ZM, Reddy AJ, Gabbita S, Nussbaum Z, Córdova CMV, Wenteler A, Weber N, Tunjic TM, et al.: **DNA-diffusion: leveraging generative models for controlling chromatin accessibility and gene expression via synthetic regulatory elements.** *BioRxiv* 2024, <https://doi.org/10.1101/2024.02.01.578352>.
- DNA-diffusion, a generative AI framework using diffusion models, can design synthetic 200 bp regulatory elements with cell type-specific activity, demonstrating that AI can generate *de novo* regulatory sequences that implicitly retain endogenous properties, including TFBS composition and context-dependent gene activation potential. This work establishes that generative models can create functional regulatory DNA *de novo* by learning from chromatin context rather than just sequence features, enabling rational design of synthetic regulatory elements tailored to specific cellular environments for therapeutic gene modulation and mammalian synthetic biology applications.
58. Routhier E, Joubert A, Westbrook A, Pierre E, Lancrey A, Cariou M, Boulé J-B, Mozziconacci J: **In silico design of DNA sequences for in vivo nucleosome positioning.** *Nucleic Acids Res* 2024, **52**:6802-6810.
- Relying on deep learning-based mutational screening, the authors design synthetic DNA sequences with programmed nucleosome positioning properties, successfully engineering kilobase-scale arrays with designer nucleosomal repeat lengths (NRL) far exceeding yeast's natural NRL. Unlike DaSilva's approach, which relies on generative diffusion models to implicitly learn regulatory activity, Routhier's framework explicitly dictates chromatin features of generated DNA sequences, such as nucleosome positioning. This demonstrates that synthetic genomics can engineer not just transcriptional output but also the underlying chromatin structural organization.
59. Nguyen E, Poli M, Durrant MG, Kang B, Katrekar D, Li DB, Bartie LJ, Thomas AW, King SH, Brixi G, et al.: **Sequence modeling and design from molecular to genome scale with Evo.** *Science* 2024, **386**:ead09336.
- See comment on ref 61 below.
60. Shao B, Yan J: **A long-context language model for deciphering and generating bacteriophage genomes.** *Nat Commun* 2024, **15**:9392.
- See comment on ref 61 below.
61. Dudek NK, Precup D: **Towards AI-designed genomes using a variational autoencoder.** *Proc R Soc B Biol Sci* 2024, **291**:20241457.
- These recent achievements shed light on emerging abilities to artificially engineer designer genomes. Evo is a genomic foundation model to generate megabase-scale functional genomic sequences. DeepGenomeVector model uses a fundamentally different approach based on a denoising variational autoencoder trained on gene-level binary vectors (gene presence/absence) to capture dependencies in genomic networks and reconstruct complete, ecologically cohesive metabolic pathways. As an intermediate approach, megaDNA implements a multiscale transformer trained on bacteriophage genomes at nucleotide resolution that generates sequences up to 96 kb and predicts regulatory elements and gene function. These reports illustrate how different architectural choices enable complementary capabilities for synthetic genome design.
62. Lee D, Yang J, Kim S: **Learning the histone codes with large genomic windows and three-dimensional chromatin interactions using transformer.** *Nat Commun* 2022, **13**:6678.
 63. Ribeiro-dos-Santos AM, Maurano MT: **Iterative improvement of deep learning models using synthetic regulatory genomics.** *Genome Res* 2025, **35**:2539-2549.
 64. Zhang D, Lam J, Blobel GA: **Engineering three-dimensional genome folding.** *Nat Genet* 2021, **53**:602-611.
 65. Piazza A, Bordelet H, Dumont A, Thierry A, Savocco J, Girard F, Koszul R: **Cohesin regulates homology search during recombinational DNA repair.** *Nat Cell Biol* 2021, **23**:1176-1186.
 66. Arnould C, Rocher V, Finoux A-L, Clouaire T, Li K, Zhou F, Caron P, Mangeot PE, Ricci EP, Mourad R, et al.: **Loop extrusion as a mechanism for formation of DNA damage repair foci.** *Nature* 2021, **590**:660-665.